Process Characterization Plan

A-Mab Drug Substance — Harvest and Clarification (Step 4) — PCP-004

Process Development / Manufacturing Science & Technology (MSAT)

2026-09-30

Table of contents

Approvals 4
Revision history 5
Abbreviations 5

1 Purpose and scope **7**
1.1 Changes in this revision 8

2 Objectives **11**

3 Related documents **12**

4 Prior knowledge and quality risk basis **13**
4.1 Unit-operation description and prior knowledge 13
4.2 Quality attributes in scope 14
4.3 Risk-based prioritization of parameters 15

5 Materials and methods **16**
5.1 Scale-down model and its qualification 16
5.2 Operation 17
5.3 Analytical methods 18
5.4 Sampling plan 18
5.5 Statistical methods 19

6 Study design **21**
6.1 Factors, ranges and study type 21
6.2 Univariate assessment 21

7 Acceptance and decision criteria **23**

8 Proven acceptable ranges (planned analysis) **25**

9 Data management and integrity **27**

10 Roles and responsibilities **28**

11 Deliverables and schedule **29**

12 Risks and assumptions **30**

13 Approvals **31**

Appendix A. Planned assessment levels **32**

Appendix B. Planned sampling and test schedule **33**

References **34**

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Approvals

Role	Function	Name (synthetic)	Signature	Date
Author	Process Development / MSAT	—	—	—
Reviewer	Analytical Development	—	—	—
Reviewer	Manufacturing Science	—	—	—
Reviewer	Statistics	—	—	—
Approver	Quality Assurance	—	—	—

Synthetic record. Signatures are not applied.

Revision history

Version	Effective date	Description of change
1.0	2025-10-20	Initial issue
2.0	2026-09-30	Reissued under INV-001: the rules of PCMP-001 Version 2.0 applied, execution dates recorded; changes listed in the document (see INV-001)

Abbreviations

Abbreviation	Meaning
ALCOA+	attributable, legible, contemporaneous, original and accurate, plus complete, consistent, enduring and available
AMV	analytical method validation
BLA	Biologics License Application
CPP	critical process parameter
CQA	critical quality attribute
DNA	deoxyribonucleic acid
ELISA	enzyme-linked immunosorbent assay
Fab	antigen-binding fragment of the antibody
Fc	crystallizable fragment of the antibody
GPP	general process parameter
HCP	host cell protein
HMW	high molecular weight species
HPLC	high-performance liquid chromatography
KPP	key process parameter
LIMS	laboratory information management system
LMW	low molecular weight species
LOQ	limit of quantitation
MSAT	Manufacturing Science and Technology
NOR	normal operating range
nrCE-SDS	non-reduced capillary electrophoresis with sodium dodecyl sulfate
NTU	nephelometric turbidity unit
PAR	proven acceptable range
QA	Quality Assurance
qPCR	quantitative polymerase chain reaction
rcf	relative centrifugal field, in multiples of the acceleration of gravity (g)
RSD	relative standard deviation
SEC	size exclusion chromatography
SOP	standard operating procedure

Abbreviation	Meaning
WC-CPP	well-controlled critical process parameter

1 Purpose and scope

This plan defines the process characterization study of harvest and clarification, Step 4 of the A-Mab drug substance process. It states which parameters are studied, how the study is run and analysed, and the criteria by which its results are judged. Version 2.0 amends Version 1.0 after the study ran (Section 1.1).

Harvest separates the antibody from the cells and the cell debris of the production culture. At commercial scale the step is run once per batch on the contents of a 15,000 L production bioreactor. The culture passes a continuous disk-stack centrifuge, a train of depth filters and a sterilizing-grade filter, in that order. The centrifuge and the filters are not expected to change a quality attribute of the antibody. Fragments form in the culture and continue to form in the clarified harvest (Section 4.2). The step determines the quality of the feed to Protein A capture. That feed should carry little intact cell material and fine debris, and the operation should break few cells, because broken cells release host cell protein and DNA into it.

In scope are the parameters of the step listed in Section 6.1. Each is studied one factor at a time on a scale-down model of the harvest train (Section 5.1), over a range that extends beyond its normal operating range (NOR). The responses are the impurity burden of the clarified harvest, its particulate content, the pressure that the filter train develops, the size-variant profile of the antibody across the step and the step yield. In the turbidity series they also include the column pressure and the cycle yield of a Protein A load challenge. The study supports the transfer of the drug substance process from the development site in Cambridge, MA to the commercial drug substance site in Grafton, WI. It forms part of the Stage 1 process design record of the commercial process (U.S. Food and Drug Administration 2011).

Five things lie outside this plan:

1. The culture conditions, which set the viability at the end of culture. Lower viability releases more host cell protein and DNA into the culture fluid before harvest. The culture conditions are characterized under the production bioreactor plan PCP-003 and are not varied here.
2. The parameters of Protein A capture, which are characterized under the Protein A plan PCP-005. In this study capture serves only as a Protein A load challenge at set-point conditions (Section 6.2).
3. The number and the arrangement of the depth filter stages, which the depth and sterile filtration procedure SOP-2006 fixes.
4. Viral clearance. No clearance is claimed for this step, and no spiking study is included (International Council for Harmonisation 2023).
5. The hold of the clarified harvest before capture. Its duration and its temperature are not parameters of this study.

1.1 Changes in this revision

Version 1.0 of this plan was approved on 2025-10-20, before the univariate study S-HAR-UNI began on 2025-11-03. Version 2.0 is issued after that study ran, under the QA investigation INV-001, which examined acceptance criteria that had been set after data review for the analysis of proven acceptable ranges. QA opened the investigation on 2026-08-03, after its review of the master report PCMR-001 Version 1.0 for BLA readiness, and closed it on 2026-09-18 with the classification major. The investigation found that the criteria used to judge each step in the proven acceptable range (PAR) analysis had been set after the data were seen and were in no approved protocol (INV-001 finding 1). No such criterion was applied to this step in Version 1.0. This revision applies the rules of the master plan PCMP-001 Version 2.0 to this step. It makes the following changes.

1. Intermediate acceptance criteria. PCMP-001 Version 2.0 derives intermediate acceptance criteria from the drug substance criteria and the platform worst cases, with no data from the studies they judge, and it verifies the platform worst cases on which they rely (PCMP-001 §7.3 and §7.5). Section 7 states how both apply to this step. The decision criteria of Version 1.0 are kept.

2. Qualification of the scale-down model. PCMP-001 §7.1 judges a scale-down model by its offset from manufacturing scale, not by a test for a difference between the scales. This replaces the Version 1.0 acceptance criterion that no performance measure should differ between the scales at $\alpha = 0.05$, with the model modified where one did. Section 5.1 states how the rule applies to the outputs of this step. The design and the procedure of the qualification are those of Version 1.0.

3. Fragments and their method. Version 1.0 listed host cell protein, residual DNA and aggregate as the quality attributes in scope. The drug substance CQA register of Version 2.0 assigns fragments to this step, so fragments are added to the attributes in scope (Section 4.2) and the nrCE-SDS method validation AMV-3020 to Table 3.2. PCMP-001 Version 2.0 sets nrCE-SDS as the method for fragments at every step, following the analytical method decision AMD-001 of 2026-03-20 (PCMP-001 §7.8). This answers INV-001 finding 6, that size exclusion chromatography (SEC) overestimates low molecular weight species where host cell protein co-elutes. Fragments were not among the responses of the univariate study of this plan, which closed before AMD-001.

4. Statistical rules. PCMP-001 §6.2 sets the significance level for the campaign at $\alpha = 0.05$, the level at which Version 1.0 of this plan already tested effects. The single-batch prediction interval of PCMP-001 §6.2 and the model-adequacy criteria of PCMP-001 §7.2 apply to response-surface models, and so not to this study (Section 5.5).

5. Classification. PCMP-001 Version 2.0 sets the rule by which each parameter is classified. An attribute is affected by a parameter when the effect of the parameter on it is significant at that level (PCMP-001 §6.2), and the class follows from the size of the effect across the proven acceptable range and the margin

of that range to the NOR (PCMP-001 §7.7). Version 1.0 left the decision logic to SOP-4001 and the evidence of the study (Section 8).

6. Execution window. Section 11 records the execution window of the univariate study S-HAR-UNI. The Version 1.0 title block carried 2026-07-24, the date on which the Version 1.0 reports were approved, in place of the approval date of this plan. The revision history now gives that approval date (INV-001 finding 5).

7. Corrections and gaps. Where Version 1.0 set no value or rule that the report needs, this plan says so and sets none. The descriptions of Version 1.0 listed below are corrected without a change to the design, and each corrected sentence is marked "(corrected, change 7)" where it stands.

    - Settling in the disc stack (Section 4.1). Version 1.0 said that a particle that has not reached the bowl wall when the fluid leaves is swept out in the centrate. In a disc stack a particle settles onto a disc, is captured there and slides outward to the bowl wall.
    - Shear in the centrifuge (Section 4.1). Version 1.0 said that shear in the feed zone and at the disc stack rises with the speed of the bowl. The shear acts in the feed zone, where the feed is accelerated to the speed of the bowl, and the flow between the discs is laminar and low in shear.
    - Passes through the feed zone (Section 5.1, Section 12). Version 1.0 said that the centrifuge recirculates part of its contents, so that the fluid passes the feed zone more than once. A continuous disk-stack centrifuge passes the feed through its feed zone once, and the cell damage that a bench model does not fully reproduce comes from the acceleration of the feed in the feed zone and from the periodic discharge of solids.
    - Field and sigma factor (Section 5.1). Version 1.0 said that a setting in g means the same thing at both scales. At each field of the series the bench feed rate is set so that its ratio of flow to sigma factor equals that of the commercial centrifuge at the same field and the commercial feed rate.
    - Turbidity and the field (Section 4.1). Version 1.0 said that the field and the filter load produce the turbidity together. A higher field lowers it by passing fewer cells and less debris, and it can raise it by breaking more cells into fine debris.
    - Depth filter load (Section 4.1). Version 1.0 tied the saturation of the media to the mass of solids per unit area and carried the load in L/m2. The load in L/m2 is the volume of centrate per unit of media area, and the solids that this volume carries set the saturation.
    - Aggregate (Section 4.2). Version 1.0 said that shear at the feed zone could unfold the antibody. An air-liquid interface, in the feed zone or in a foaming transfer, can unfold it, and shear breaks cells.
    - Ranges and levels (Section 4.3, Section 6.1, Section 6.2). Version 1.0 said that every range studied is wider than the NOR at both ends, that the depth filter load range reaches well beyond its upper NOR limit, and that every planned level is a value of the control strategy. For post-clarification turbidity only the upper edge of the range studied extends beyond the NOR, the depth filter load range reaches beyond its upper NOR limit, and three of the five levels are values of the control strategy. A note to Table 6.1 adds that

post-clarification turbidity is measured, not set, in the centrifugation (rcf) and depth filter load series.

- Size-variant samples (Section 5.4). Version 1.0 said that the intermediate sample points would not locate a change in the size-variant profile. A sample of the centrate would locate it, and the centrate is not tested for size variants.
- Analysis (Section 5.3, Section 5.5, Section 12). Version 1.0 said in its method section that a change smaller than the method can resolve would not be reported as an effect, that the decisions of its criteria rest on the size of an effect against the precision of its method, and that residual DNA lies close to the limit of quantitation of the qPCR. Such a change is reported as detected and not material, as Section 5.5 sets out, the decisions also rest on significance and on an absolute limit, and the residual DNA of a clarified harvest lies far above that limit.
- Presumed results (Section 2, Section 4.2, Section 8). Version 1.0 said that the study would confirm that the NOR lies inside the PAR, that the drug substance criteria are met by the clearance downstream, that the NOR will be shown nested inside the PAR, and what the step contributes to the control strategy. This plan states each as an intention.

The design, the sampling, the procedures and the decision rules of Version 1.0 are otherwise kept in its words, except the descriptions corrected under change 7.

2 Objectives

The study has five objectives.

1. Qualify a scale-down model of the commercial harvest train and show that it represents the centrifugation and depth filtration performance of that train (Section 5.1).
2. Establish the effect of the centrifugal field, over 6,000 g to 12,000 g, on the clarity of the centrate and on the host cell protein and DNA released by cell breakage.
3. Establish the effect of depth filter load, over 80 L/m2 to 160 L/m2, on the particulate content of the clarified harvest and on the pressure the filter train develops.
4. Establish whether the step alters the size-variant profile of the antibody, and establish the effect of post-clarification turbidity, over 1 NTU to 20 NTU, on the material presented to capture.
5. Define a proven acceptable range for each parameter, determine whether the normal operating range lies inside the proven acceptable range, and provide the evidence on which each parameter will be classified under the parameter classification procedure SOP-4001 (corrected, change 7).

Classification is an outcome of the study and is reported in PCR-004.

3 Related documents

Table 3.1 lists the documents that scope this study and the documents it feeds. The risk assessment RA-001 decides which parameters are studied and by which type of study (Section 4.3), and the paired report PCR-004 reports the study that this plan sets out.

Table 3.1: Related documents.

Document ID	Title	Relationship
PTP-001	Process Transfer Plan (A-Mab Drug Substance)	Parent — transfer scope
RA-001	Pre-Characterization Process Risk Assessment (A-Mab Drug Substance)	Parent — risk basis for study scope
PCMP-001	Process Characterization Master Plan (A-Mab Drug Substance)	Parent — master plan
PCR-004	Process Characterization Report (Harvest and Clarification (Step 4))	Sibling — paired document
PCMR-001	Process Characterization Master Report (A-Mab Drug Substance)	Rolls up into
INV-001	Investigation Report (Acceptance criteria for proven-acceptable-range analysis set after data review)	Reason for this revision

Table 3.2 lists the procedures and the validated methods that govern the study.

Table 3.2: Procedures and method validations referenced in this document.

Type	Reference	Title
Standard operating procedure	SOP-1001	Qualification of Scale-Down Models
	SOP-2005	Harvest by Continuous Disk-Stack Centrifugation
	SOP-2006	Clarification by Depth and Sterile Filtration
	SOP-4001	Process-Parameter Classification and Control-Strategy Definition
Analytical method validation	AMV-3015	Turbidity by Nephelometry (NTU)
	AMV-3012	Host-Cell Protein ELISA
	AMV-3014	Residual DNA (qPCR)
	AMV-3011	Size-Variants (SEC-HPLC)
	AMV-3020	Purity (non-reduced CE-SDS)

4 Prior knowledge and quality risk basis

4.1 Unit-operation description and prior knowledge

At commercial scale the culture is fed continuously to a disk-stack centrifuge. Cells and debris settle under the centrifugal field, at a velocity that rises with the field and with the difference between their density and the density of the medium. In the disc stack a particle settles onto a disc, is captured there and slides outward to the bowl wall, from which the solids are discharged at intervals (corrected, change 7). A higher feed rate shortens the time the fluid spends in the stack, so that more of the small particles escape capture. The centrate therefore still carries the fine debris and the colloidal material that are too small to settle in the time available.

The depth filter train removes that material. Its media are positively charged, so they adsorb part of the DNA and fine debris as well as retaining particles by size. The filtrate then passes the sterilizing-grade filter into the harvest vessel, where it is held until capture.

A higher centrifugal field clarifies more completely, because cells and debris settle faster and fewer escape capture. It can also break more cells. The culture is accelerated to the speed of the bowl in the feed zone, and the shear there rises with that speed, while the flow between the discs is laminar and low in shear (corrected, change 7). Broken cells release host cell protein and DNA into the centrate, and they leave fine debris that is harder to capture. The set-point balances clarification against cell breakage.

A depth filter has a finite capacity for the solids and colloids that the centrifuge passes. As retained solids fill the void space of the matrix and occupy its charged sites, the pressure drop across the filter rises and more fine material passes. At the extreme, cells and debris are forced through the pores. How far the media are saturated depends on the volume of centrate that each unit of media area has received and on the solids that this volume carries. For this reason the parameter is expressed as the volume of centrate loaded per unit of media area, in L/m2 (corrected, change 7).

Post-clarification turbidity measures the fine particulate and colloidal material that the filters passed. It is not a set parameter. A higher centrifugal field lowers it, because fewer cells and less debris reach the filters, and it can also raise it, because more broken cells leave fine debris (corrected, change 7). It rises as the filter load approaches the capacity of the media. It is monitored on every batch as an in-process control. It is listed among the parameters of the step because the particulate and colloidal load that it measures reaches the Protein A column, where it fouls the bed and shortens the working life of the resin.

Harvest by continuous centrifugation and depth filtration is the platform recovery operation at Novacyte Biologics. It was used for three earlier antibodies (X-Mab, Y-Mab and Z-Mab), and the equipment, the grade of the filter media and the control scheme are unchanged for A-Mab. The mechanisms above are therefore expected to hold for A-Mab, and the platform experience gives the approximate position of the operating point. Host cell protein and DNA enter harvest at a concentration that rises as the viability of the culture falls, because cells that died before harvest have already released their contents into the medium. Viability at the end of culture is a property of the A-Mab culture and is characterized under the production bioreactor plan PCP-003. The ranges in this plan are therefore established on A-Mab material.

The A-Mab case study assumes, for simplicity, that harvest has no impact on product quality, and it notes that this might not hold in a real process (CMC Biotech Working Group 2009). This plan keeps that assumption, and the criteria of PCMP-001 Version 2.0 rest on it (Section 7). The study tests it where cell breakage could release host cell protein and DNA, and where air-liquid interfaces could add aggregate.

4.2 Quality attributes in scope

Table 4.1 lists the four quality attributes in scope, with the criticality and the drug substance criterion of each. The Tool #1 score is the product of the impact score and the uncertainty score of the ranking tool of the A-Mab case study (CMC Biotech Working Group 2009).

Table 4.1: Quality attributes in scope at harvest, with the drug substance criteria.

Category	Quality attribute	Drug substance criterion	Criticality	Tool #1
process impurity	Host Cell Protein (HCP)	≤ 100 ng/mg	M-H	36
	Residual DNA	≤ 0.001 ng/dose	VL	6
size variant	Aggregates (HMW)	≤ 5 % HMW (SEC)	H	60
	Fragments (LMW, clipped species)	≤ 2 % (nrCE-SDS)	L	12

The drug substance criteria are those of the drug substance CQA register (PCMP-001 Table 3.1), which takes them from the A-Mab case study.
For residual DNA a dose is 10 mg/kg (A-Mab case study) for a reference adult of 70 kg, an assumption of this campaign, so 700 mg of antibody.
Criticality is very low (VL), low (L), moderate (M), high (H) or very high (VH), and M-H is moderate to high.

Host cell protein, DNA and aggregate enter harvest in the culture fluid and leave it in the clarified harvest. The step is not credited with clearing them. The depth filter media adsorb part of the DNA (Section 4.1), but no clearance is claimed for it. Aggregate is reduced mainly at cation exchange. The three are in scope because shear and air-liquid interfaces in the centrifuge and the filters can break cells, which adds host cell protein and DNA, and because an air-liquid interface in the feed zone or

in a foaming transfer can unfold the antibody and add aggregate (corrected, change 7). Aggregate has a higher criticality than host cell protein, but an effect on it is less expected, because the antibody meets such an interface only briefly. Residual DNA has very low criticality. It is measured because DNA fragments and nucleosomes foul depth filters and Protein A resin, so it describes the condition of the feed even where its own criterion is far from binding.

Fragments are clipped antibody, mostly cleaved in or near the hinge into Fab and Fab+Fc species. Clipping starts in the reactor, where proteases released by dead and lysing cells act on the secreted antibody. It continues in the clarified harvest, because clarification removes the cells but not the proteases already in solution, and its rate rises with temperature. A longer culture ends at lower viability and carries more protease into harvest. The master plan PCMP-001 sets no criterion for fragments at this step, and they are judged at the downstream steps (PCMP-001 Table 7.1). Fragments are not a response of the study in this plan.

The drug substance criteria in Table 4.1 are to be met by the clearance that the purification train delivers after this step, and they are not in-process limits for the clarified harvest (corrected, change 7). The criteria by which this step is judged are those of Section 7.

4.3 Risk-based prioritization of parameters

The parameters of this study and the study type of each follow the pre-characterization risk assessment RA-001. RA-001 ranked each parameter on its potential to affect a quality attribute and on its potential to interact with another parameter, and it assigned a study type from the combined ranking. It brought 3 parameters of this step into scope and assigned each of them to univariate study (Table 6.1). No parameter was ranked high enough to require a multivariate design, so no screening or response-surface design is planned for this step (PCMP-001 §2). The other operating conditions are held constant (Section 5.2). The feed rate follows the ratio of flow to sigma factor, the depth filtration runs at constant flux, and the temperature is held at the commercial set-point.

The depth filter media adsorb part of the DNA, and the share that they take up can change with the filter load. Protein A capture and the two ion exchange steps clear the host cell protein and DNA that cell breakage releases (Section 4.2). A poorly chosen setting is therefore expected to load capture more heavily without changing the drug substance.

The range studied extends beyond the NOR at both ends for centrifugation (rcf) and depth filter load. For post-clarification turbidity the lower edge of the range studied is the lower limit of the NOR, and only the upper edge extends beyond it (corrected, change 7). A range that stops at the NOR limits cannot show where the operation begins to fail, and an excursion beyond a NOR limit would then be a deviation with no data behind it. The wider ranges also cover settings that the receiving site may need if its equipment differs from the development equipment.

5 Materials and methods

The materials and methods are those of Version 1.0. Where this amendment applies a rule of PCMP-001 Version 2.0, the text names it.

5.1 Scale-down model and its qualification

The characterization runs will be executed on a scale-down model of the harvest train, qualified under the scale-down model procedure SOP-1001 before the first characterization run.

The model reproduces the commercial operation in three parts, and each part is matched on the quantity that drives its own mechanism and not on the size of the equipment. Matching on volume or on bowl diameter alone would leave the shear at the feed zone, the settling in the bowl and the loading of the filter media free to differ between the scales. Cell damage in the feed zone will be reproduced in a bench shear device operated at the energy dissipation rate calculated for the commercial feed zone. Sedimentation will be reproduced in a laboratory centrifuge operated at the same relative centrifugal field as the commercial bowl and at a matched ratio of flow to sigma factor. At each field of the series the bench feed rate will be set so that its ratio of flow to sigma factor equals that of the commercial centrifuge at the same field and the commercial feed rate (corrected, change 7). Depth filtration will be reproduced with disks of the commercial filter media, run at the same flux and to the same load per unit area, so that a setting expressed in L/m2 carries across in the same way.

Qualification will compare the model with the commercial train at the set-point condition on the same culture. The comparison will cover centrate turbidity, solids removal, host cell protein and DNA in the clarified harvest, filter pressure at the end of the load, and step yield. Replicate runs will be performed at each scale. At commercial scale each replicate is a separate batch.

PCMP-001 §7.1 replaces the Version 1.0 acceptance criterion of this comparison (change 2). It judges each output by its offset, the mean of the scale-down runs at set-point minus the mean of the commercial batches, with its two-sided 90 % confidence interval. Step yield, centrate turbidity, solids removal and filter pressure have no criterion at this step, so their offsets would only be recorded. Host cell protein and DNA are judged against the impurity release criterion (criterion 1 of Section 7). PCMP-001 §7.1 decides the qualification of an output with a criterion on the response-surface model of the step, which this step does not have. This plan therefore sets no qualification rule, and PCR-004 states its approach. PCMP-001

§7.1 also asks each step plan to name the number of runs at each scale and the manufacturing-scale data set. Version 1.0 named neither, and this amendment, issued after the study, does not add them.

The turbidity of the clarified harvest, which the clarification limit of 10 NTU judges (criterion 2), is not in the comparison, so its offset is unknown, not zero. The size variants and the column pressure and the cycle yield of the Protein A load challenge are not in the comparison either.

A continuous disk-stack centrifuge passes the feed through its feed zone once. The cell damage that a bench model does not fully reproduce comes from the acceleration of the feed in the feed zone, which the shear device matches only in its energy dissipation rate, and from the periodic discharge of solids (corrected, change 7). The model therefore represents the field and the residence time well and that cell damage less well, and it is expected to under-represent lysis rather than to over-represent it. This study uses the model to bound ranges and to establish the direction and the approximate size of each effect. It does not use the model to predict the absolute impurity concentration of a commercial batch.

5.2 Operation

Each characterization run will be executed as a single pass of the whole harvest through the model train, in the order the commercial operation runs it.

The culture will be taken from the production bioreactor scale-down model at the end of culture, at the conditions characterized under the production bioreactor plan PCP-003. Culture from the qualified bioreactor model matters here, because a culture that ends at lower viability enters harvest with more host cell protein already released into the medium and with more fragile cells. Both raise the impurity concentration of the clarified harvest, whatever the centrifuge does. Where the volume allows, one culture will be split between the levels of a parameter, so that the levels of a series are compared on the same material.

The culture will be fed to the centrifuge model at the field setting for the run and at the matched ratio of flow to sigma factor. The centrate will be collected, sampled, and passed over the depth filter train at constant flux to the load per unit area for the run, and then over a sterilizing-grade filter into the harvest container. Temperature will be held at the commercial set-point throughout. The time between the end of culture and the end of clarification will be held constant across the runs, since a longer hold on unclarified culture allows further lysis and would confound the comparison the study is making.

Every parameter not varied in a run will be held at its set-point, and the set-points are those given in Table 6.1. Runs will be executed against the procedures listed in Table 3.2.

5.3 Analytical methods

Four of the validated methods in Table 3.2 report the responses of the study. Turbidity by nephelometry (AMV-3015) measures the fine particulate and colloidal material the filters pass, and it is the in-process measurement by which clarified harvest is released to capture. Host cell protein by ELISA (AMV-3012) and residual DNA by qPCR (AMV-3014) measure the impurity burden of the clarified harvest. Size variants by SEC-HPLC (AMV-3011) measure the high molecular weight content of the antibody before and after the step. Version 1.0 named no method for the antibody concentration on which step yield rests, and this amendment names none.

Table 5.1 gives the validated performance of the turbidity, host cell protein and DNA methods. The performance of AMV-3011 is given in its own validation report.

Table 5.1: Validated performance of the turbidity, host cell protein and DNA methods.

Reference	Method	Precision, RSD (%)	LOQ	Accuracy (%)
AMV-3015	Turbidity by Nephelometry (NTU)	4	0.5 NTU	98
AMV-3012	Host-Cell Protein ELISA	9.5	1 ng/mg	95
AMV-3014	Residual DNA (qPCR)	15	1 pg/mg	92

Precision is the relative standard deviation (RSD) of a single result.

Samples for host cell protein and for DNA will be assayed in one analytical campaign per parameter series, with the same reagent lots where the schedule allows. A trend across the levels of a parameter should not be confounded by a change of reagent lot part way through the series.

5.4 Sampling plan

Samples will be drawn at four points in every run. The points, the methods applied at each and the purpose of each are listed in Appendix B (Table B.1).

The sample of unclarified culture is the reference for the whole study. Impurity released by the operation can only be seen as a difference from what entered it, so that sample will be drawn from the production bioreactor scale-down model at the end of culture, immediately before the feed to the centrifuge starts, and it will be assayed in the same campaign as the samples that follow it. The centrate sample separates the two stages of the operation. It shows what the centrifuge passed and therefore what the depth filter was asked to remove, which is the quantity that the depth filter load runs on centrate from 6,000 g and from 12,000 g vary (Section 5.5). The sample after depth filtration and the sample after sterilizing-grade filtration are both clarified harvest, and the second of them is the material capture receives.

Each level will be sampled once at each point. The set-point condition will be run in replicate across the execution of the study, and those replicates provide the estimate of run to run variability against which every effect is tested. Version 1.0 set no number of replicates, and this amendment sets none.

Samples for size variants will be drawn at the first and the last point only. The antibody meets air-liquid interfaces only briefly in the operation, so the step is not expected to change the size-variant profile (Section 4.2). The centrate is not tested for size variants, so a change found at the last point could not be placed at the centrifuge or at the filters (corrected, change 7).

5.5 Statistical methods

Each response will be analysed against the parameter that was varied, by linear regression on the levels run, with the replicates at the set-point condition providing an independent estimate of error. Effects will be tested at $\alpha = 0.05$, the significance level set for the campaign (PCMP-001 §6.2). Where the relationship is not linear over the range, a quadratic term may be fitted, and the fit will be reported with the terms that were needed rather than with the terms that were planned. Version 1.0 named no test for curvature, and this amendment adds none.

Statistical significance and practical significance will be reported separately. A significant slope will be reported together with the change it predicts across the normal operating range of that parameter, and that change will be compared with the precision of the method that measured the response (Section 5.3). An effect smaller than the method can resolve will be reported as detected and not material, with the comparison shown.

No adequacy threshold is set for the regressions. The decisions of Section 7 rest on the significance of a change, on its size against the resolution of the method, or on an absolute limit, and not on the fit of a predictive model (corrected, change 7). Each regression will therefore be reported with its coefficient of determination and its residual standard error and will be judged on those.

Results below the limit of quantitation of a method will be reported as such and excluded from the regression, and the number excluded will be given with the fit.

The design is one factor at a time. It cannot determine an interaction between two parameters, where the effect of one parameter on a response depends on the level of the other. One interaction is expected on mechanistic grounds, since a higher centrifugal field passes fewer solids to the depth filter and so changes the load that filter has to handle. That interaction will be bracketed and not estimated. The depth filter load series will be run on centrate produced at 6,000 g and again on centrate produced at 12,000 g, so that the filter is challenged with both extremes of the solids load it can receive. The bracket bounds the filter response between the two solids loads the centrifuge can deliver, and it yields no coefficient for the interaction, so none will be reported.

No response-surface model will be fitted for this step and no Monte-Carlo propagation of a fitted model will be performed, because no multivariate design is run here. A range reported from this study therefore holds with the other parameters at their set-points, and it says nothing about two of them moving away from their set-points together.

6 Study design

The study design is that of Version 1.0.

6.1 Factors, ranges and study type

Table 6.1 gives the parameters, their set-points, the ranges to be studied and the NORs. The set-points and the NORs are those against which the study was planned.

Table 6.1: Parameters in scope and the ranges to be studied, each studied one factor at a time.

Parameter	Unit	Set-point	Range studied	NOR
Centrifugation (rcf)	g	9,000	6,000–12,000	8,000–10,000
Depth filter load	L/m2	120	80–160	100–140
Post-clarification turbidity[a)]	NTU	5	1–20	1–10

[a)] Measured, not set, in the centrifugation (rcf) and depth filter load series. Its set-point is a level of its own series.

The ranges serve different purposes. Centrifugation (rcf) runs from 6,000 to 12,000 g, so that the series covers both the lower fields, at which clarification is less complete, and the higher fields, at which more cells may break (Section 4.1). Depth filter load runs to 160 L/m2, which loads the media beyond the upper NOR limit of 140 L/m2 and past the point at which the pressure drop is expected to rise (corrected, change 7). Post-clarification turbidity is the response of the other two parameters, and the range in its row is the range over which turbidity will be presented to capture, not a range of settings that will be dialled in at the centrifuge.

The set-point column is the condition at which every parameter not being varied will be held, and it is the condition at which the replicate runs of Section 5.4 will be executed.

6.2 Univariate assessment

Each parameter will be assessed at five levels. They are the lower edge of the range studied, the lower limit of the normal operating range, the set-point, the upper limit of the normal operating range and the upper edge of the range studied, and they are

listed in Appendix A (Table A.1). Three of the five are values that the control strategy uses, so the study reports directly on the limits it is meant to justify (corrected, change 7). For post-clarification turbidity the lower edge of the range studied and the lower limit of the NOR coincide, so its series has four distinct levels.

Runs will be executed in randomized order within each parameter series. A drift in the culture, in the equipment or in an assay over the execution of a series would otherwise be read as an effect of the parameter. Replicates at the set-point condition will be distributed across a series rather than run together, for the same reason.

The centrifugation (rcf) series will be run first, because its outcome supplies the centrate produced at 6,000 g and at 12,000 g on which the depth filter load series is run (Section 5.5). The depth filter load series will follow. The turbidity series will be run last, because the material for it is produced by the two series before it. Clarified harvest at a target turbidity will be prepared by blending clarified harvest with retained centrate to the target turbidity for the level, which allows the turbidity presented to capture to be varied independently of the settings that produced it. Blended material carries the particle size distribution of the centrate it was made from, and that is not identical to the distribution a depth filter at the end of its capacity would pass.

The turbidity series will be loaded onto a scale-down Protein A column held at the set-point conditions of the Protein A plan PCP-005, as a Protein A load challenge. The responses are the pressure developed across the column over the load and the yield of the cycle. Version 1.0 set no criterion for these two responses, and this amendment sets none. No conclusion about the capture step will be drawn from these runs, and the characterization of capture itself is reported in PCR-005.

7 Acceptance and decision criteria

The master plan PCMP-001 Version 2.0 derives an intermediate acceptance criterion for a step output by carrying the drug substance criterion back through the platform worst case of every downstream step, with no margin (PCMP-001 §7.3). PCMP-001 Table 7.1 sets no intermediate acceptance criterion for the clarified harvest, because the derivation credits harvest with no change in any attribute, so the criteria of the production bioreactor output pass through it unchanged. For the same reason the verification of platform worst cases (PCMP-001 §7.5) has no value to check at this step. Those criteria therefore rest on the assumption that harvest changes no attribute (Section 4.1), which the impurity release criterion (1) and the product quality criterion (3) below test.

The decision criteria below are those of Version 1.0, in its words. The impurity release criterion (1) and the product quality criterion (3) compare results within a run, and the clarification criterion (2) is an absolute limit. No result of the studies under this plan was used to set or change these criteria.

1. Impurity release is the first criterion. Host cell protein and residual DNA in the clarified harvest will be compared with the same measurements on the unclarified culture that produced it. A level passes when no increase is detected at $\alpha = 0.05$, or when a detected increase is smaller than the resolution of the method that measured it (Section 5.3). A setting that releases more impurity than that lies outside the acceptable range of its parameter.
2. Clarification is the second criterion and the absolute one. Post-clarification turbidity of the clarified harvest will be compared with the upper limit of its normal operating range, 10 NTU. A centrifuge setting or a filter load that delivers clarified harvest above that value has failed to clarify, whatever the impurity results show.
3. Product quality is the third. High molecular weight content will be compared before and after the step, and a level passes when no increase is detected at $\alpha = 0.05$.
4. Step yield is reported but is not gated. Yield will be given for every level and compared with the set-point replicates. No numeric yield limit is set in this plan, because none follows from a product quality requirement at this step. A loss of yield at a level will be reported with its cause and will be weighed in the choice of the operating range. It will not on its own place a level outside the proven acceptable range.

A parameter meets the criteria over a range when every level in that range passes the first three criteria. The proven acceptable range of the parameter is then that range, bounded at the levels tested (Section 8). Where a level fails, the proven acceptable

range is bounded at the last level that passed, and the failure should be investigated before the range is reported.

A failure at a level inside the normal operating range means the operating range in force is not supported by the data, and it should be raised to Manufacturing Science and to Quality Assurance before the study continues, so that the normal operating range or the control strategy can be changed under SOP-4001. Deviations from this plan during execution will be recorded, investigated and reported in PCR-004 with their impact on the conclusions, whether or not the affected runs are retained.

8 Proven acceptable ranges (planned analysis)

PCMP-001 §6.4 computes proven acceptable ranges from the response-surface model of a step, once with the other parameters held at their set-points and once with them drawn at random around their set-points. That method needs a response-surface model (Section 5.5). PCMP-001 §4 states that the range of a parameter studied univariately is supported by its own data, so the proven acceptable range of each parameter of this step will be determined from the levels tested, by the rule of Version 1.0 below.

A proven acceptable range is the range of a parameter over which the step has been shown to meet its criteria, with every other parameter held at its set-point (International Council for Harmonisation 2009). A range established by varying one parameter at a time holds while the other parameters are at their set-points (Section 5.5), and no combination of two parameters away from their set-points is claimed for this step.

A proven acceptable range is also bounded by the levels tested. The analysis will not extrapolate beyond the edges given in Table 6.1. Where every level of a parameter meets the criteria, the proven acceptable range will be reported as the whole range studied and not as anything wider.

Version 1.0 gave no size for the resolution of a method in the impurity release criterion, no threshold for judging the regressions on their coefficient of determination and residual standard error, no rule for the proven acceptable range of the post-clarification turbidity series, and no rule for how the proven acceptable range of depth filter load follows from its two series, on centrate from 6,000 g and from 12,000 g. This amendment sets none of them, and PCR-004 states its approach to each.

For each parameter, PCR-004 will give the range studied, the criterion applied, the response that came closest to that criterion, and the proven acceptable range that follows. It will determine whether the normal operating range lies inside the proven acceptable range (corrected, change 7). Where the two coincide at an edge, that will be stated, because a normal operating limit sitting on the edge of the proven range leaves no margin for the ordinary variability of the commercial operation.

No design space is claimed for this step. A design space is a multivariate region, and this study runs no multivariate design. What the study is to provide to the control strategy is a proven acceptable range for each parameter, the position of the normal operating range against it, and an in-process control on the turbidity of the clarified harvest (corrected, change 7).

The classification of each parameter under SOP-4001 will be made in PCR-004 on the evidence this study produces, and it will roll up into PCMR-001 with the other steps of the train.

PCMP-001 Version 2.0 sets the rule for that classification (change 5). An attribute is affected by a parameter when the effect of the parameter on it is significant at the significance level set for the campaign (PCMP-001 §6.2). A critical process parameter (CPP) and a well-controlled critical process parameter (WC-CPP) both affect a CQA, and they differ in the risk that the parameter falls outside its acceptable range in operation. A key process parameter (KPP) affects process performance and no CQA, and a general process parameter (GPP) affects neither (PCMP-001 §7.7). Neither PCMP-001 nor SOP-4001 sets a numerical threshold for a practically relevant effect. PCR-004 will therefore state, for each parameter, the size of each effect across the proven acceptable range and the margin of that range to the normal operating range, and it will give the reason for each class from those two quantities.

9 Data management and integrity

Data from this study will be recorded, reviewed and retained under the site data governance procedures, and the ALCOA+ principles apply to every record the study produces.

Runs will be executed against a pre-approved batch record for the scale-down train. Each entry will be made at the time it is observed and signed by the person who made it. Analytical results will be acquired and stored in the validated laboratory information management system (LIMS), with the audit trail enabled and reviewed as part of result approval. Equipment used to set or to measure a parameter of this study will be within its calibration at the time of use, and the calibration status will be recorded with the run. The first and the last day of execution of each study will be recorded with its data, as PCMP-001 Version 2.0 requires, and PCMP-001 Table 9.1 records them for the univariate study S-HAR-UNI.

Raw data, the analysis script and the outputs of that script will be retained together, so that the tables in PCR-004 can be regenerated from the raw data without a manual step. Any change to a recorded value will be made as a corrected entry, with the reason and the original value visible. The analysis will be verified by a second person in Statistics, and the report will be reviewed by the functions in the approval record.

10 Roles and responsibilities

Process Development is responsible for the study. Table 10.1 lists the functions that take part and what each is accountable for.

Table 10.1: Functions and responsibilities for the harvest characterization study.

Function	Responsibility
Process Development / MSAT	Owns this plan, executes the runs on the scale-down model, interprets the results and writes PCR-004
Analytical Development	Releases the methods, performs the assays and reports method performance with the results
Manufacturing Science (receiving site)	Confirms the commercial operation the scale-down model represents and reviews the ranges proposed
Statistics	Agrees the analysis before execution, performs it, and verifies the analysis as second person
Quality Assurance	Approves this plan and PCR-004, and approves the disposition of any deviation

Two of these functions are engaged before execution. The analysis of Section 5.5 will be agreed with Statistics before the first run. The commercial operation the scale-down model represents will be confirmed with Manufacturing Science at the receiving site before qualification, since the model is matched to that operation and not to the development one (International Council for Harmonisation 2008).

11 Deliverables and schedule

The deliverable of this study is the Process Characterization Report for Harvest and Clarification, PCR-004. It will report the qualification of the scale-down model, the results of each parameter series, the proven acceptable ranges, the classification of each parameter and any deviation from this plan.

The study will run in four phases, in order. The scale-down model will be qualified first, and no characterization run will be executed until that qualification has been reviewed and accepted. The centrifugation (rcf) series will follow, then the depth filter load series on centrate from the extremes of the centrifugation range, then the turbidity series on material prepared from both. Analysis of a series may begin once the last analytical result of that series has been approved, and the report will be issued when all three series have been analysed.

Version 1.0 fixed no dates. It held the schedule in the transfer plan PTP-001, where it was coordinated with the other characterization studies of the campaign and with the availability of the bioreactor scale-down model. The study chronology of the campaign gives the execution window of the univariate study S-HAR-UNI, from its first day of execution on 2025-11-03 to its last on 2025-11-21 (PCMP-001 Table 9.1).

12 Risks and assumptions

Five things could weaken the conclusions of this study, and each carries a mitigation or a bound (Version 1.0). A sixth item states the assumption on which the study rests.

1. The scale-down model is the largest of them. A continuous disk-stack centrifuge is the hardest unit in this process to represent at bench scale, for the reason given in Section 5.1, and the model is expected to under-represent lysis (corrected, change 7). The mitigation is that the study reports effects and ranges rather than absolute impurity levels, and that the qualification of Section 5.1 compares the model with the commercial train on the responses that matter here.
2. The study assumes the culture is representative. A culture that ends at lower viability carries more debris and more broken cells into harvest, which raises both the solids load on the centrifuge and the host cell protein released before the run begins. Culture will therefore be taken from the qualified bioreactor scale-down model at its set-point condition, and the viability and the cell density of every culture used will be recorded with the run and reported in PCR-004.
3. Material supply constrains what can be compared within a series. One culture may not provide enough volume for all five levels of a parameter at the scale of the model, so a series may span more than one culture. Where it does, the set-point replicates run on each culture provide the reference for the levels run on that culture, and the analysis will fit the culture as a block.
4. The blended material of the turbidity series carries the particle size distribution of centrate and not that of a saturated depth filter (Section 6.2). The turbidity range will therefore be reported as a range of turbidity delivered to capture, and no claim will be made about the specific material a filter at the end of its capacity would pass.
5. Assay variability bounds the smallest effect the study can see (Table 5.1).
6. The assumption that harvest has no impact on product quality (Section 4.1). The impurity release criterion and the product quality criterion of Section 7 test it, for host cell protein and DNA released by cell breakage and for aggregate added at air-liquid interfaces.

13 Approvals

This plan takes effect when it has been approved by every function in the approval record at the front of this document. Execution of the study did not begin before the approval of Version 1.0. Any change to the plan after approval will be made as a revision to this document and approved by the same functions. A change made during execution that is not a revision to this plan is a deviation, and it is handled as Section 7 describes.

Appendix A. Planned assessment levels

Table A.1 gives the planned levels of each series, with the material each series uses. As Section 5.5 describes, the depth filter load series will be run twice, once on centrate from each end of the range studied for centrifugation (rcf).

Table A.1: Planned levels of each series.

Series	Material	Level	Setting
Centrifugation (rcf) series	Culture from the production bioreactor scale-down model	Lower edge of range studied	6,000 g
		Lower limit of NOR	8,000 g
		Set-point	9,000 g
		Upper limit of NOR	10,000 g
		Upper edge of range studied	12,000 g
Depth filter load series	Centrate produced at 6,000 g	Lower edge of range studied	80 L/m2
		Lower limit of NOR	100 L/m2
		Set-point	120 L/m2
		Upper limit of NOR	140 L/m2
		Upper edge of range studied	160 L/m2
	Centrate produced at 12,000 g	Lower edge of range studied	80 L/m2
		Lower limit of NOR	100 L/m2
		Set-point	120 L/m2
		Upper limit of NOR	140 L/m2
		Upper edge of range studied	160 L/m2
Post-clarification turbidity series	Clarified harvest blended with retained centrate	Lower edge of range studied and lower limit of NOR	1 NTU
		Set-point	5 NTU
		Upper limit of NOR	10 NTU
		Upper edge of range studied	20 NTU

In every series the set-point condition will also be run in replicate, distributed across the series. Version 1.0 set no number of replicates.

Appendix B. Planned sampling and test schedule

Table B.1 lists the four sample points of every run, the methods applied at each and what each point is for. The methods are those of Table 3.2.

Table B.1: Planned sample points and the methods applied at each.

Sample point	Methods	Purpose
Unclarified culture, drawn before the feed to the centrifuge starts	Turbidity, host cell protein, residual DNA, size variants	Reference against which impurity release and any size-variant change are judged
Centrate, after centrifugation	Turbidity, host cell protein, residual DNA	What the centrifuge passed, and the solids load the depth filter receives
Filtrate, after depth filtration	Turbidity, host cell protein, residual DNA	What the depth filter media retained and adsorbed
Clarified harvest, after sterilizing-grade filtration	Turbidity, host cell protein, residual DNA, size variants	The material presented to Protein A capture

Turbidity by nephelometry (AMV-3015), host cell protein by ELISA (AMV-3012), residual DNA by qPCR (AMV-3014), size variants by SEC-HPLC (AMV-3011).

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